ChatGPT said:

Laboratory Practice III – Oral / Viva Preparation

Savitribai Phule Pune University (SPPU), Fourth Year Computer Engineering, 2019 Pattern – Course 410246.

Your uploaded syllabus contains three subjects:

1. Group A – Design and Analysis of Algorithms (DAA)
2. Group B – Machine Learning (ML)
3. Group C – Blockchain Technology (BT)

You need the first 5 practicals from each subject, including basic theory, full forms, important definitions, working principles, and simple viva answers.

I'll keep the explanations in easy English so you can understand them and speak them confidently during your practical oral examination.

Important full forms for oral

Short form	Full form
DAA	Design and Analysis of Algorithms
ML	Machine Learning
BT	Blockchain Technology
SPPU	Savitribai Phule Pune University
AI	Artificial Intelligence
DS	Dataset
CPU	Central Processing Unit
RAM	Random Access Memory
TLE	Time Limit Exceeded
T.C.	Time Complexity
S.C.	Space Complexity
DP	Dynamic Programming
B&B	Branch and Bound
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
GD	Gradient Descent
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
AI	Artificial Intelligence
NN	Neural Network
EDA	Exploratory Data Analysis
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
ETH	Ether
EVM	Ethereum Virtual Machine
dApp	Decentralized Application
P2P	Peer-to-Peer
PoW	Proof of Work

PoS	Proof of Stake
DLT	Distributed Ledger Technology
BTC	Bitcoin
ETH	Ethereum

Group A – Design and Analysis of Algorithms

Practical 1: Fibonacci Series – Recursive and Non-Recursive

Basic theory

What is Fibonacci series?

The Fibonacci series is a sequence in which each number is the sum of the previous two numbers.

Example:

`0, 1, 1, 2, 3, 5, 8, 13, 21...`

Mathematical formula:

$$F(n) = F(n-1) + F(n-2)$$

Base conditions:

$$F(0) = 0, \quad F(1) = 1$$

What is recursion?

Recursion is a programming technique in which a function calls itself to solve a smaller problem.

Example: Fibonacci recursive function calls itself for `n-1` and `n-2`.

What is a non-recursive method?

A non-recursive method uses loops such as `for` or `while` instead of calling itself.

Difference

Recursive	Non-recursive
Function calls itself	Uses loops
More function-call overhead	Less overhead
Simple to understand mathematically	Generally more efficient
Exponential time for basic Fibonacci	Linear time

Complexity

Method	Time complexity	Space complexity
Recursive Fibonacci	$O(2^n)$	O(n)
Non-recursive Fibonacci	O(n)	O(1)

Viva questions

Q1. What is an algorithm?

An algorithm is a finite sequence of steps used to solve a problem.

Q2. What is time complexity?

Time complexity measures how the execution time of an algorithm grows with input size.

Q3. What is space complexity?

Space complexity measures the amount of memory required by an algorithm.

Q4. What is the base condition in recursion?

It is the condition that stops recursive calls.

Q5. Which Fibonacci method is more efficient?

The non-recursive method is generally more efficient because it avoids repeated calculations.

Practical 2: Huffman Encoding Using Greedy Strategy

Basic theory

What is Huffman encoding?

Huffman encoding is a lossless data compression technique that assigns variable-length binary codes to characters based on their frequencies.

Frequently occurring characters get shorter codes, while less frequent characters get longer codes.

What is a greedy algorithm?

A greedy algorithm selects the best available choice at each step to achieve an overall solution.

Working principle

1. Calculate the frequency of every character.
2. Create a leaf node for each character.
3. Select the two nodes with the lowest frequencies.
4. Combine them into a new node.
5. Repeat until only one node remains.
6. Assign binary codes using 0 and 1.

Example

Suppose the frequencies are:

Character	Frequency
A	5
B	9
C	12
D	13
E	16
F	45

The character `F` has the highest frequency and will generally receive a short code.

Important terms

- **Prefix code:** No code is the prefix of another code.
- **Lossless compression:** Original data can be recovered exactly.
- **Frequency:** Number of occurrences of a character.
- **Binary code:** A code made using 0 and 1.

Complexity

Using a min-priority queue:

- Time complexity: O(n log n)
- Space complexity: O(n)

Viva questions

Q1. What is the main purpose of Huffman encoding?

To reduce the size of data using lossless compression.

Q2. Why is it called a greedy algorithm?

Because it repeatedly selects the two minimum-frequency nodes.

Q3. What is a prefix code?

A code in which no codeword is the prefix of another codeword.

Q4. Is Huffman encoding lossless or lossy?

It is lossless.

Q5. What data structure is commonly used?

A min-priority queue or min-heap.

Practical 3: Fractional Knapsack Using Greedy Method

Basic theory

The Knapsack problem is an optimization problem in which we select items to maximize total profit without exceeding the capacity of a bag.

In the fractional knapsack problem, we can take a fraction of an item.

Formula

Profit-to-weight ratio:

$$R_i = \frac{P_i}{W_i}$$

Where:

- P_i = Profit of item
- W_i = Weight of item
- R_i = Profit per unit weight

Working principle

1. Calculate profit-to-weight ratio.
2. Sort items in descending order of ratio.
3. Select the item with the highest ratio.
4. If the complete item fits, take it.
5. Otherwise, take the required fraction.
6. Stop when the bag is full.

Example

Bag capacity = 50 kg.

Item	Weight	Profit	Ratio
A	10	60	6
B	20	100	5
C	30	120	4

Take:

- Item A: 10 kg, profit 60
- Item B: 20 kg, profit 100
- 20 kg of Item C: profit 80

Total profit = 240.

Complexity

- Time complexity: O(n log n), due to sorting.
- Space complexity: O(n), depending on implementation.

Viva questions

Q1. Why is fractional knapsack greedy?

It selects items based on the highest profit-to-weight ratio.

Q2. Can an item be divided?

Yes, an item can be taken partially.

Q3. What is the objective?

To maximize total profit within the given capacity.

Q4. Is greedy always optimal for fractional knapsack?

Yes.

Q5. What is the difference between fractional and 0-1 knapsack?

Fractional knapsack allows partial items, while 0-1 knapsack allows only complete items or no item.

Practical 4: 0-1 Knapsack Using Dynamic Programming

Basic theory

The 0-1 Knapsack problem is an optimization problem where each item can either be selected completely or not selected.

We cannot take a fraction of an item.

What is dynamic programming?

Dynamic Programming (DP) is a technique that solves a problem by dividing it into smaller overlapping subproblems and storing their results.

Formula

Let $DP[i][w]$ represent the maximum profit using the first `i` items with capacity `w`.

If the item weight is greater than capacity:

$$DP[i][w] = DP[i-1][w]$$

Otherwise:

$$DP[i][w] = \max(DP[i-1][w], P_i + DP[i-1][w - W_i])$$

Working principle

1. Initialize the DP table.
2. Consider each item one by one.
3. Check whether the item fits.
4. Compare taking the item with not taking it.
5. Store the maximum profit.
6. The final answer is in `DP[n][W]`.

Complexity

- Time complexity: O(nW)
- Space complexity: O(nW)

Here `n` is the number of items and `W` is the bag capacity.

Viva questions

Q1. What does 0-1 mean?

Each item is either selected (1) or not selected (0).

Q2. Why is DP used?

It avoids repeated calculations by storing previous results.

Q3. Can fractional items be taken?

No.

Q4. What is the difference between greedy and DP?

Greedy makes locally optimal choices, while DP compares subproblem solutions to find the optimum.

Q5. What is the optimal substructure property?

The optimal solution can be constructed from optimal solutions to smaller subproblems.

Practical 5: N-Queens Using Backtracking

Basic theory

The N-Queens problem is a chessboard problem in which we place N queens on an N × N chessboard so that no two queens attack each other.

What is backtracking?

Backtracking is an algorithmic technique that builds a solution step by step and removes a choice when it leads to an invalid solution.

Rules

No two queens can share:

- The same row
- The same column
- The same diagonal

Working principle

1. Start from the first row.
2. Place a queen in a safe column.
3. Move to the next row.
4. Check whether the position is safe.
5. If no safe position exists, backtrack.
6. Continue until all N queens are placed.

Example

For N = 4, one valid arrangement is:

Plain text

```
. Q . .
. . . Q
Q . . .
. . Q .
```

Complexity

- Worst-case time complexity: O(N!)
- Space complexity: O(N), excluding the board representation.

Viva questions

Q1. What is backtracking?

It is a method of trying possible solutions and undoing choices that fail.

Q2. Why is N-Queens a backtracking problem?

Because a queen is removed and another position is tried when a conflict occurs.

Q3. What is a safe position?

A position where the queen is not attacked by any previously placed queen.

Q4. How many queens are placed on an N × N board?

Exactly N queens.

Q5. What is the use of backtracking?

It is used in puzzles, constraint satisfaction problems, and combinatorial problems.

Group A practicals 1–5 are based on the supplied SPPU syllabus.

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Group B – Machine Learning

Practical 1: Uber Fare Prediction Using Linear Regression and Random Forest

Basic theory

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every situation.

What is regression?

Regression is a supervised machine learning technique used to predict continuous numerical values.

Example: Predicting the price of an Uber ride.

What is preprocessing?

Data preprocessing is the process of cleaning and preparing raw data before applying a machine learning algorithm.

Common preprocessing techniques:

- Removing missing values
- Removing duplicate records
- Encoding categorical data
- Scaling numerical values
- Identifying outliers

What is an outlier?

An outlier is a data point that is significantly different from other observations.

Example: An Uber fare of ₹50,000 for a short ride may be an outlier.

What is correlation?

Correlation measures the relationship between two variables.

Its value lies between -1 and +1.

Value	Meaning
+1	Perfect positive correlation
0	No linear correlation
-1	Perfect negative correlation

Linear Regression

Linear Regression predicts a continuous value using a linear relationship.

Formula:

$$y = mx + c$$

Where:

- `y` = Predicted output
- `x` = Input
- `m` = Slope
- `c` = Intercept

Random Forest Regression

Random Forest is an ensemble learning algorithm that combines multiple decision trees to make a prediction.

For regression, the final output is generally the average of the predictions from all trees.

Model evaluation

Metric	Meaning
R^2 Score	Measures how well the model explains variation
RMSE	Measures average prediction error magnitude
MSE	Mean squared prediction error

$$RMSE = \sqrt{\frac{1}{n}\sum(y - \hat{y})^2}$$

Lower RMSE indicates smaller prediction errors.

Viva questions

Q1. What is the target variable in Uber fare prediction?

The fare amount.

Q2. What is the difference between regression and classification?

Regression predicts numerical values, while classification predicts categories.

Q3. What is Random Forest?

It is an ensemble algorithm that combines multiple decision trees.

Q4. What is the purpose of data preprocessing?

To improve data quality and prepare it for machine learning.

Q5. Which model is better: Linear Regression or Random Forest?

The model with better evaluation performance on the test data is preferred. The result depends on the dataset.

Q6. What does RMSE indicate?

It indicates the magnitude of prediction errors in the same unit as the target variable.

Practical 2: Email Spam Detection Using KNN and SVM

Basic theory

What is classification?

Classification is a supervised machine learning technique used to assign data to predefined categories.

In email spam detection:

- Normal State = Not Spam
- Abnormal State = Spam

What is binary classification?

Binary classification is a classification problem with exactly two classes.

Example: Spam or Not Spam.

KNN – K-Nearest Neighbors

KNN is a supervised learning algorithm that classifies a new data point based on the majority class among its K nearest neighbors.

Working principle of KNN

1. Choose the value of K.
2. Calculate distances between the new point and training points.
3. Find the K nearest points.
4. Count the classes of those neighbors.
5. Assign the majority class.

What is SVM?

SVM stands for Support Vector Machine.

It is a supervised learning algorithm that finds a decision boundary separating different classes.

Important SVM terms

- **Hyperplane:** A decision boundary that separates classes.
- **Support vectors:** Data points closest to the decision boundary.
- **Margin:** Distance between the hyperplane and the nearest data points.
- **Kernel:** A function used to handle nonlinear classification.

KNN vs SVM

KNN	SVM
Distance-based	Boundary-based
Uses nearest neighbors	Uses support vectors
Simple to understand	Can handle complex boundaries
Prediction can be slower on large datasets	Training may be computationally expensive

Performance analysis

Common metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Viva questions

Q1. What is spam?

Spam is an unwanted or unsolicited email.

Q2. What is K in KNN?

K is the number of nearest neighbors considered for classification.

Q3. What is SVM?

SVM is a supervised machine learning algorithm used for classification and regression.

Q4. What is a support vector?

A data point closest to the separating hyperplane.

Q5. Which algorithm is lazy learning?

KNN is called a lazy learning algorithm because it delays generalization until prediction.

Q6. Why is feature scaling important in KNN?

Because KNN uses distances, and features with large numerical ranges can dominate the distance calculation.

Practical 3: Bank Customer Churn Prediction Using Neural Network

Basic theory

What is customer churn?

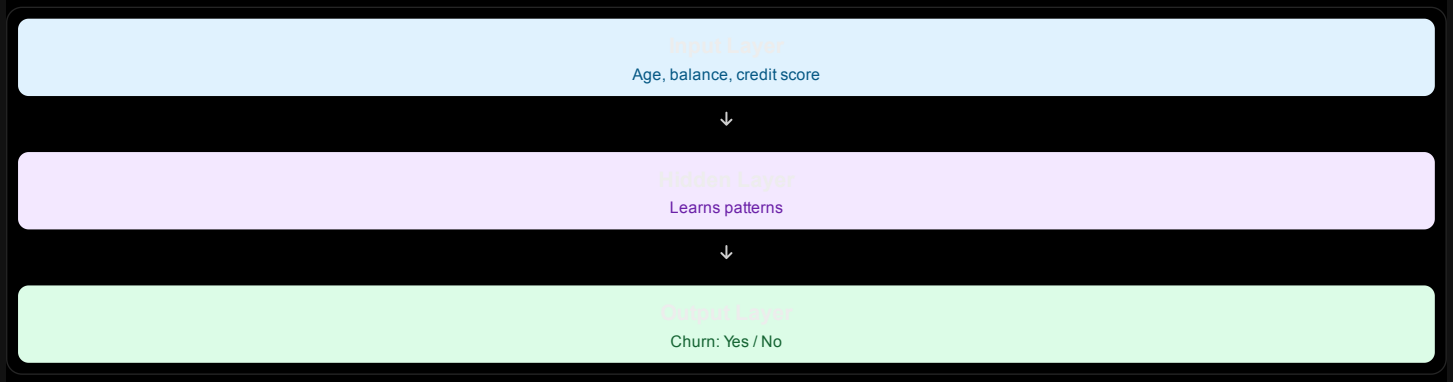
Customer churn means a customer stops using the services of a company or bank.

Example: A customer closes their bank account.

What is a Neural Network?

A Neural Network is a machine learning model inspired by the structure of the human brain. It consists of interconnected artificial neurons.

Basic structure



Important terms

Neuron: Basic computational unit of a neural network.

Weights: Parameters that determine the importance of inputs.

Bias: An additional parameter used to shift the output.

Activation function: Function that introduces non-linearity into the network.

Epoch: One complete pass through the training dataset.

Backpropagation: Method used to calculate gradients and update weights during training.

What is normalization?

Normalization scales numerical features to a common range.

One common formula is:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Train-test split

The dataset is divided into:

- Training data: Used to train the model.
- Testing data: Used to evaluate the model on unseen data.

Confusion matrix

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

Viva questions

Q1. What is customer churn prediction?

It predicts whether a customer will leave a company or bank.

Q2. What is an epoch?

One complete pass through the training dataset.

Q3. What is backpropagation?

It calculates gradients of the loss and helps update network weights.

Q4. Why do we normalize data?

To bring features to a common scale and improve model training.

Q5. What is an activation function?

A function that transforms a neuron's output and helps the network learn nonlinear patterns.

Q6. What is the output of this experiment?

A prediction indicating whether the customer will leave or stay.

Practical 4: Gradient Descent Algorithm

Basic theory

What is Gradient Descent?

Gradient Descent is an optimization algorithm used to minimize a function by repeatedly moving in the direction of the negative gradient.

It is commonly used to optimize machine learning models.

What is a local minimum?

A local minimum is a point where a function has a lower value than nearby points.

Given function

$$y = (x + 3)^2$$

Starting point:

$$x = 2$$

The derivative is:

$$\frac{dy}{dx} = 2(x + 3)$$

Gradient descent formula

$$x_{\text{new}} = x_{\text{old}} - \alpha f'(x)$$

Where:

- `x` = Current point
- `α` = Learning rate
- `f'(x)` = Gradient of the function

Example

If learning rate $\alpha = 0.1$:

At $x = 2$:

$$f'(2) = 2(2 + 3) = 10$$

$$x_{\text{new}} = 2 - 0.1(10) = 1$$

The next point moves closer to the minimum at x = -3.

Important terms

Learning rate: Controls the step size during optimization.

Gradient: Derivative of a function.

Convergence: When the algorithm approaches a stable solution.

Iteration: One update of the current parameter.

Viva questions

Q1. What is the purpose of Gradient Descent?

To minimize a function or loss value.

Q2. What is the learning rate?

It controls how much the parameter changes during each update.

Q3. What happens if the learning rate is too high?

The algorithm may overshoot the minimum or fail to converge.

Q4. What happens if the learning rate is too low?

Convergence may become very slow.

Q5. What is the minimum of the given function?

For $y = (x + 3)^2$, the minimum occurs at x = -3, where y = 0.

Practical 5: KNN on Diabetes Dataset

Basic theory

KNN is a supervised learning algorithm used to classify data based on the nearest data points.

In this practical, KNN is applied to a diabetes dataset to predict whether a person has diabetes.

Dataset features

Typical diabetes dataset features include:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function

- Age

The target is generally:

- 0 = No diabetes
- 1 = Diabetes

Distance formula

Euclidean distance:

$$d = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

It calculates the distance between two data points.

Working principle

1. Load the diabetes dataset.
2. Separate features and target.
3. Divide data into training and testing sets.
4. Normalize or scale the features.
5. Choose the value of K.
6. Train the KNN classifier.
7. Predict the test data.
8. Calculate performance metrics.

Performance metrics

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Error Rate:

$$Error = 1 - Accuracy$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

Viva questions

Q1. Why is KNN used in diabetes prediction?

To classify patients into diabetes and non-diabetes categories based on their features.

Q2. What is Euclidean distance?

It is the straight-line distance between two points.

Q3. What is precision?

Precision measures how many predicted positive cases are actually positive.

Q4. What is recall?

Recall measures how many actual positive cases are correctly identified.

Q5. What is a confusion matrix?

It is a table used to compare actual and predicted classifications.

Q6. Why is scaling important in this experiment?

Because features such as glucose, age, and insulin have different ranges, and KNN depends on distance.

Group B practicals 1–5 are based on the supplied SPPU syllabus.

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Group C – Blockchain Technology

Practical 1: Installation of MetaMask and Study Spending Ether per Transaction

Basic theory

What is Blockchain?

Blockchain is a distributed digital ledger that stores transactions in blocks linked together using cryptographic techniques.

What is MetaMask?

MetaMask is a cryptocurrency wallet that allows users to manage Ethereum accounts and interact with blockchain applications.

What is Ether?

Ether (ETH) is the native cryptocurrency of the Ethereum blockchain.

Important terms

Term	Meaning
Wallet	Software used to manage blockchain accounts
Transaction	An operation recorded on the blockchain
Gas	Unit measuring computational work on Ethereum
Gas Price	Price paid per unit of gas
Gas Limit	Maximum gas units allowed for a transaction
Transaction Fee	Fee paid to process a transaction
Testnet	Blockchain network used for testing
Mainnet	The main production blockchain network

Transaction fee formula

$$\text{Fee} = \text{Gas Used} \times \text{Effective Gas Price}$$

For example:

Gas used = 21,000 units

Gas price = 10 Gwei

$$21000 \times 10 = 210000 \text{ Gwei}$$

$$= 0.00021 \text{ ETH}$$

Working principle

1. Install MetaMask.
2. Create or import a wallet.

3. Select an Ethereum network.
4. Obtain test ETH on a testnet for testing.
5. Send a transaction.
6. Check the transaction hash.
7. Observe gas used and transaction fee.

Viva questions

Q1. What is MetaMask?

MetaMask is a cryptocurrency wallet used to manage Ethereum accounts and interact with dApps.

Q2. What is Ether?

Ether is the native cryptocurrency of Ethereum.

Q3. What is gas?

Gas measures the computational work required to execute a transaction.

Q4. What is a transaction hash?

It is a unique identifier used to locate a blockchain transaction.

Q5. What is the difference between mainnet and testnet?

Mainnet is used for real transactions, while testnet is used for testing.

Q6. Is gas the same as Ether?

No. Gas measures computational work, while Ether is the cryptocurrency used to pay transaction fees.

Practical 2: Create Your Own Wallet Using MetaMask

Basic theory

A blockchain wallet is a tool used to manage cryptographic keys and interact with blockchain accounts.

A wallet does not directly store cryptocurrency. The cryptocurrency is recorded on the blockchain.

Important terms

Public Address: An address used to receive cryptocurrency.

Private Key: A secret cryptographic key used to authorize transactions.

Seed Phrase: A sequence of recovery words used to restore a wallet.

Cryptocurrency: A digital asset secured using cryptographic techniques.

Digital Signature: A cryptographic mechanism used to prove authorization of a transaction.

Public key vs private key

Public Key / Address	Private Key
Used to receive funds	Used to authorize transactions
Can be shared	Must remain secret
Identifies an account	Controls access to the account

Wallet creation steps

1. Install MetaMask.
2. Create a new wallet.
3. Set a password.
4. Securely save the recovery phrase.
5. Confirm the recovery phrase.
6. Open the wallet.

7. Copy the public address.

Viva questions

Q1. What is a cryptocurrency wallet?

A tool used to manage keys and interact with blockchain accounts.

Q2. What is a public address?

It is an address used to receive cryptocurrency.

Q3. What is a private key?

A secret key used to authorize transactions.

Q4. What is a seed phrase?

A recovery phrase used to restore a wallet.

Q5. Can we share the seed phrase?

No. Anyone who knows the seed phrase may be able to control the wallet.

Q6. What is a digital signature?

A cryptographic proof that a transaction was authorized by the holder of the private key.

Practical 3: Smart Contract for Bank Account Operations

Basic theory

What is a smart contract?

A smart contract is a program deployed on a blockchain that executes according to predefined rules.

In this practical, a smart contract provides:

- Deposit money
- Withdraw money
- Show balance

What is Solidity?

Solidity is a programming language used to write smart contracts for Ethereum-compatible blockchains.

Important terms

Smart Contract: A program deployed on a blockchain.

Solidity: A smart contract programming language.

Remix IDE: A browser-based IDE used to write, compile, deploy, and test Solidity smart contracts.

EVM: Ethereum Virtual Machine, the execution environment for Ethereum smart contracts.

Mapping: A Solidity data structure that stores key-value pairs.

Basic Solidity concepts

Keyword	Meaning
`contract`	Defines a smart contract
`function`	Defines an operation
`payable`	Allows a function to receive Ether
`msg.sender`	Address of the transaction sender
`msg.value`	Ether amount sent with a transaction
`mapping`	Stores key-value associations

`require()`	Checks a condition and reverts if false

Working principle

1. Write the smart contract in Solidity.
2. Compile it using Remix IDE.
3. Deploy it on an Ethereum test network.
4. Deposit Ether into the contract.
5. Withdraw Ether.
6. Check the balance.
7. Observe transaction fees.

Viva questions

Q1. What is a smart contract?

A program deployed on a blockchain that executes according to predefined rules.

Q2. What is Solidity?

Solidity is a programming language used to write Ethereum smart contracts.

Q3. What is Remix IDE?

Remix is an online IDE for developing and testing smart contracts.

Q4. What does `payable` mean?

It allows a function or address to receive Ether.

Q5. What is `msg.sender`?

It represents the address that initiated the current transaction or call.

Q6. What is `msg.value`?

It represents the amount of Ether sent with the transaction, measured in wei.

Q7. What is wei?

Wei is the smallest denomination of Ether.

$$1\ \text{ETH} = 10^{18}\ \text{wei}$$

Practical 4: Student Data Using Solidity – Structures, Arrays, and Fallback

Basic theory

In this practical, we create a Solidity smart contract to store student information using structures, arrays, and fallback functions.

What is a structure?

A structure (`struct`) is a user-defined data type that groups different variables together.

Example:

A student can have:

- Roll number
- Name
- Marks

These variables can be stored in a single structure.

What is an array?

An array is a collection of elements of the same data type.

Example:

```solidity
uint[] marks;
```

This represents an array of unsigned integers.

What is a fallback function?

A fallback function is a special function in Solidity that is executed when a call does not match any declared function, or when certain conditions involving calldata are met.

It can also receive Ether if declared payable.

Important terms

Term	Meaning
`struct`	Groups multiple variables
`array`	Stores multiple elements
`fallback()`	Special function for unmatched calls
`gas`	Computational cost
`transaction fee`	Cost of executing a transaction
`deployment`	Publishing a smart contract to a blockchain

What is gas value?

Gas value represents the amount of computational work consumed by a transaction.

What is transaction fee?

The transaction fee is the amount paid to execute a transaction or deploy a contract.

Working principle

1. Create a Solidity contract.
2. Define a student structure.
3. Create an array of student records.
4. Write functions to add and retrieve student data.
5. Implement a fallback function.
6. Compile and deploy the contract.
7. Observe gas consumption and transaction fees.

Viva questions

Q1. What is a struct?

A user-defined data type that groups multiple variables.

Q2. What is an array?

A collection of elements of the same data type.

Q3. What is a fallback function?

A special function that handles unmatched function calls or certain calldata conditions.

Q4. What is gas?

Gas measures the computational work performed by a smart contract.

Q5. Why does smart contract deployment require gas?

Because deployment requires computation and blockchain storage.

Q6. What is the difference between gas and transaction fee?

Gas measures computational work, while the transaction fee is the payment for that work.

Practical 5: Survey Report on Types of Blockchains and Real-Time Use Cases

Basic theory

A blockchain is a distributed ledger in which transactions are stored in blocks linked through cryptographic hashes.

Blockchain is designed to provide transparency, data integrity, and distributed record keeping.

Types of blockchain



1. Public Blockchain

A public blockchain is open to anyone who wants to participate.

Examples: Bitcoin, Ethereum.

Use case: Cryptocurrency transactions.



2. Private Blockchain

A private blockchain is controlled by a single organization.

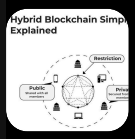
Use case: Internal company records and supply chain management.



3. Consortium Blockchain

A consortium blockchain is controlled by a group of organizations.

Use case: Banking networks and inter-organizational data sharing.



4. Hybrid Blockchain

A hybrid blockchain combines public and private blockchain features.

Use case: Systems that require both controlled access and public verification.

Important blockchain terms

Decentralization: Control is distributed across multiple participants instead of one central authority.

Distributed Ledger: A database shared and synchronized across multiple participants.

Consensus Mechanism: A method used by network participants to agree on the state of the blockchain.

Cryptographic Hash: A fixed-length output generated from input data.

Immutability: The difficulty of changing recorded information after it has been accepted into the blockchain.

Node: A computer participating in a blockchain network.

Real-time use cases

Application	Use
Cryptocurrency	Digital payments
Supply Chain	Tracking goods
Healthcare	Medical record integrity
Voting	Transparent election records
Banking	Financial transactions
Digital Identity	Identity verification
Charity	Transparent donation tracking

Viva questions

Q1. What are the four main types of blockchain?

Public, Private, Consortium, and Hybrid.

Q2. What is decentralization?

Distribution of control among multiple participants.

Q3. What is a distributed ledger?

A shared database maintained by multiple participants.

Q4. What is a consensus mechanism?

A method used by blockchain participants to agree on the valid state of the ledger.

Q5. What is a hash?

A fixed-length output generated from input data using a hash function.

Q6. What is immutability?

The property that makes recorded blockchain data difficult to alter.

Q7. What is the difference between public and private blockchain?

Public blockchains are open to participation, while private blockchains restrict participation to authorized users.

Q8. What is a consortium blockchain?

A blockchain controlled by a group of organizations.

Q9. What is a blockchain use case in healthcare?

Maintaining integrity and traceability of medical records.

Q10. What is a dApp?

A decentralized application that uses blockchain or smart contracts as part of its backend.

Group C practicals 1–5 are based on the supplied SPPU syllabus.

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Final 15 Practical Viva Revision

These are the most important one-line answers to memorize before the oral examination.

Question	Answer
What is an algorithm?	A finite sequence of steps to solve a problem.
What is recursion?	A function calling itself.
What is a greedy algorithm?	An algorithm that selects the best available choice at each step.
What is dynamic programming?	A technique that stores solutions to subproblems.
What is backtracking?	Trying choices and undoing invalid choices.
What is Machine Learning?	Learning patterns from data to make predictions.
What is supervised learning?	Learning from labeled data.
What is regression?	Predicting continuous numerical values.
What is classification?	Predicting categories.
What is clustering?	Grouping similar data points without predefined labels.
What is KNN?	Classification based on nearest neighbors.
What is SVM?	A model that finds a decision boundary between classes.
What is Gradient Descent?	An optimization method that minimizes a function.
What is a neural network?	A model made of interconnected artificial neurons.

What is a blockchain?	A distributed ledger of linked blocks.
What is a smart contract?	A program deployed on a blockchain.
What is Solidity?	A language for writing smart contracts.
What is MetaMask?	A wallet for managing Ethereum accounts.
What is Ether?	Ethereum's native cryptocurrency.
What is gas?	A measure of computational work on Ethereum.

Oral exam tip: For every practical, prepare these four things first: the aim, the basic theory, the algorithm or working principle, and the time complexity or evaluation metric. If the examiner asks about your code, explain what the input is, what the program does, and what the output means.

The material above covers the first five practicals of all three groups from your supplied syllabus.

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